

THE HERITAGE SCHOOL
FIRST TERM UNIT TEST, 2026-27
MATHEMATICS · CLASS 12 · CHAPTERS 1 - 5

Maximum Marks: 40

Time Allotted: One and a half hours

Instructions to Candidates

- You are allowed an additional 10 minutes for only reading the paper.
- You must NOT start writing during reading time.
- The Question Paper consists of **eleven questions** and **four sections**: Sections A, B, C and D. **All questions are compulsory.**
- **Section A** consists of very short answer questions of **1 mark** each.
- **Section B** consists of short answer questions of **2 marks** each.
- **Section C** consists of moderately long answer questions of **3 marks** each.
- **Section D** consists of long answer questions of **5 marks** each.
- Internal choices have been provided in **two questions** in Section B, **two questions** in Section C and **one question** in Section D.
- While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
- The intended marks for questions or parts of questions are given in brackets [].
- All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
- Mathematical tables are provided.

Syllabus covered: Relations and Functions; Inverse Trigonometric Functions; Matrices; Determinants; Continuity and Differentiability.

Not examined in this paper: Applications of Derivatives, Integrals, Application of Integrals and Linear Programming — these are in the First Term but outside this test.

SECTION A - 10 MARKS

Question 1 [10 × 1]

In sub-parts (i) to (x), choose the correct option or answer the question as instructed.

- (i) The number of reflexive relations on the set $A = \{a, b, c\}$ is:
 - (a) 64
 - (b) 512
 - (c) 8
 - (d) 6
- (ii) If $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = 2x - 3$, find $f^{-1}(5)$.
- (iii) Write the principal value of $\sec^{-1}(-2)$.
- (iv) Find the value of $\tan^{-1}(\tan 3\pi/4)$.
- (v) A matrix has 12 elements. Write the number of possible orders it can have.
- (vi) If A is a square matrix such that $A^2 = A$, find the value of $(I + A)^2 - 3A$.
- (vii) Find the value(s) of x for which

$$\begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix}$$

- (viii) Find k so that the points $(1, 2)$, $(3, k)$ and $(5, 6)$ are collinear.
- (ix) The function $f(x) = \tan x$ is discontinuous at:
 - (a) $x = (2n + 1)\pi/2$, $n \in \mathbb{Z}$
 - (b) $x = n\pi$, $n \in \mathbb{Z}$
 - (c) no point of $\mathbb{R}$
 - (d) every real number
- (x) If $y = e^{\sin x}$, find dy/dx .

SECTION B - 8 MARKS

Question 2 [2]

Let $A = \{1, 2, 3, 4\}$ and let R be the relation on A defined by $R = \{(a, b) : b = a + 1\}$. Show that R is neither reflexive, nor symmetric, nor transitive.

Question 3 [2]

Find the value of $\sin(\pi/3 - \sin^{-1}(-1/2))$.

OR

Write $\tan^{-1}x/(\sqrt{a^2 - x^2})$, $|x| < a$, in its simplest form.

Question 4 [2]

If $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, show that $A^2 = -I$ and hence find A^4 .

Question 5 [2]

If $y = \sqrt{x + \sqrt{x}}$, find dy/dx .

OR

If $x = a(\theta + \sin \theta)$ and $y = a(1 - \cos \theta)$, find dy/dx .

SECTION C - 12 MARKS**Question 6 [3]**

Solve for x : $\tan^{-1}(x - 1)/(x - 2) + \tan^{-1}(x + 1)/(x + 2) = \pi/4$.

OR

Prove that $\cos^{-1}4/5 + \cos^{-1}12/13 = \cos^{-1}33/65$.

Question 7 [3]

If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$,
verify that $(AB)' = B'A'$.

Question 8 [3]

If $y = (\tan^{-1}x)^2$, show that $(1 + x^2)^2(d^2y)/(dx^2) + 2x(1 + x^2)dy/dx = 2$.

OR

If $y = x^{\sin x} + (\sin x)^x$, find dy/dx .

Question 9 [3]

If $y = a e^{2x} + b e^{-x}$, where a and b are constants, show that
 $(d^2y)/(dx^2) - dy/dx - 2y = 0$

SECTION D - 10 MARKS**Question 10 [5]**

Solve the following system of equations by the matrix method:

$$x + y + z = 6, \quad y + 3z = 11, \quad x - 2y + z = 0$$

OR

Using properties of determinants, prove that

$$\begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix} = (1 - x^3)^2$$

Question 11 [5]

Let $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$ and let R be the relation on A given by $R = \{(a, b) : |a - b| \text{ is a multiple of } 4\}$. Show that R is an equivalence relation, and find the set of all elements related to 1.

ANSWER KEY & MARKING NOTES

Unit Test, 40 marks. **Do not read this section before attempting the paper.** Answers were checked against a computer algebra system; the notes give the decisive step rather than a full solution.

Section A

Q	Answer
1 (i)	(a) 64. The three pairs (a,a), (b,b), (c,c) must be present; each of the remaining $3^2 - 3 = 6$ ordered pairs is free, giving $2^6 = 64$.
1 (ii)	4. Solving $2x - 3 = 5$ gives $x = 4$.
1 (iii)	$2\pi/3$. The range of $\sec^{-1}$ is $[0, \pi] - \{\pi/2\}$, and $\sec(2\pi/3) = -2$.
1 (iv)	$-\pi/4$. $3\pi/4 \notin (-\pi/2, \pi/2)$, but $\tan 3\pi/4 = -1$, so the value is $\tan^{-1}(-1)$.
1 (v)	6. The orders are $1 \times 12, 2 \times 6, 3 \times 4, 4 \times 3, 6 \times 2$ and 12×1 — one for each divisor of 12.
1 (vi)	I. $(I + A)^2 = I + 2A + A^2 = I + 3A$, and subtracting $3A$ leaves I.
1 (vii)	$x = \pm\sqrt{3}$. The left side is -18 and the right is $2x^2 - 24$, so $2x^2 = 6$.
1 (viii)	$k = 4$. Collinearity needs the determinant of $[[1,2,1],[3,k,1],[5,6,1]]$ to vanish, which gives $-4k + 16 = 0$.
1 (ix)	(a). $\tan x = \sin x / \cos x$ is undefined exactly where $\cos x = 0$, that is at the odd multiples of $\pi/2$.
1 (x)	$e^{\sin x} \cos x$, by the chain rule.

Section B

Q	Answer
2	$R = \{(1,2), (2,3), (3,4)\}$. Not reflexive: $(1,1) \notin R$. Not symmetric: $(1,2) \in R$ but $(2,1) \notin R$. Not transitive: $(1,2)$ and $(2,3) \in R$ but $(1,3) \notin R$.
3	$\sin^{-1}(-1/2) = -\pi/6$, so the angle is $\pi/3 + \pi/6 = \pi/2$, and $\sin \pi/2 = \mathbf{1}$.
3 (OR)	Put $x = a \sin \theta$, so $\sqrt{(a^2 - x^2)} = a \cos \theta$ and the argument becomes $\tan \theta$. Hence the expression is θ , i.e. $\sin^{-1}(x/a)$.
4	$A^2 = [[-1, 0], [0, -1]] = -I$. Therefore $A^4 = (A^2)^2 = (-I)^2 = I$.
5	$dy/dx = (2\sqrt{x} + 1)/(4\sqrt{x} \cdot \sqrt{(x + \sqrt{x})})$, by the chain rule applied twice.
5 (OR)	$dx/d\theta = a(1 + \cos \theta)$ and $dy/d\theta = a \sin \theta$, so $dy/dx = (\sin \theta)/(1 + \cos \theta) = \mathbf{\tan(\theta/2)}$.

Section C

Q	Answer
6	Combining the two terms and setting the result equal to $\tan^{-1}1$ gives $((x - 1)(x + 2) + (x + 1)(x - 2))/((x - 2)(x + 2) - (x - 1)(x + 1)) = 1$. Cross-multiplying and simplifying reduces to $4x^2 = 2$, so $x = \pm\mathbf{1/\sqrt{2}}$ (both roots satisfy the original equation).
6 (OR)	Let $A = \cos^{-1}4/5$ and $B = \cos^{-1}12/13$, so $\cos A = 4/5$, $\sin A = 3/5$, $\cos B = 12/13$ and $\sin B = 5/13$. Then $\cos(A + B) = 48/65 - 15/65 = 33/65$, and $A + B$ lies in $[0, \pi]$.
7	$AB = [[4, 6], [10, 12]]$, so $(AB)' = [[4, 10], [6, 12]]$. $B' = [[2, 1], [0, 3]]$ and $A' = [[1, 3], [2, 4]]$, so $B'A' = [[4, 10], [6, 12]]$. The two agree.
8	$dy/dx = (2 \tan^{-1}x)/(1 + x^2)$, so $(1 + x^2)dy/dx = 2 \tan^{-1}x$. Differentiating this again and multiplying through by $(1 + x^2)$ gives the result.
8 (OR)	Put $u = x^{\sin x}$ and $v = (\sin x)^x$ and use logarithmic differentiation on each. $dy/dx = x^{\sin x}[\cos x \log x + (\sin x)/x] + (\sin x)^x[\log(\sin x) + x \cot x]$.
9	$dy/dx = 2a e^{2x} - b e^{-x}$ and $(d^2y)/(dx^2) = 4a e^{2x} + b e^{-x}$. Substituting, $(4a e^{2x} + b e^{-x}) - (2a e^{2x} - b e^{-x}) - 2(a e^{2x} + b e^{-x}) = 0$ identically.

Section D

Q	Answer
10	$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & -2 & 1 \end{bmatrix}$ has $ A = -9 \neq 0$, so A^{-1} exists and $X = A^{-1}B$ gives $\mathbf{x} = \mathbf{1}, \mathbf{y} = \mathbf{2}, \mathbf{z} = \mathbf{3}$. Check: $1 + 2 + 3 = 6 \checkmark, 2 + 9 = 11 \checkmark, 1 - 4 + 3 = 0 \checkmark$.
10 (OR)	Apply $R_1 \rightarrow R_1 + R_2 + R_3$: every entry of the first row becomes $1 + x + x^2$. Take that factor out, then use $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$ to reach a form whose expansion is $(1 + x + x^2)^2(1 - x)^2 = (\mathbf{1 - x^3})^2$.
11	Reflexive: $ a - a = 0 = 4 \times 0$. Symmetric: $ a - b = b - a $, so if one is a multiple of 4 so is the other. Transitive: if $a - b = 4k$ and $b - c = 4m$ then $a - c = 4(k + m)$. Hence R is an equivalence relation. Elements related to 1 are those differing from 1 by a multiple of 4: {1, 5, 9} .